Hugo Marquerie 11/02/2026

Lem diferencia productos leq suma diferencias

Lema 1. *Sean* $\{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subset \overline{\mathbb{D}}$*. Entonces,* $\left|\prod_{k=1}^n a_k - \prod_{k=1}^n b_k\right| \leqslant \sum_{k=1}^n |a_k - b_k|$.

***Demostración*:** Lo probaremos por inducción en n. Para $n = 1$, el resultado es trivial. Supongamos que el resultado se cumple para n. Consideramos $\{a_k\}_{k=1}^{n+1}, \{b_k\}_{k=1}^{n+1} \subset \overline{\mathbb{D}}$. Entonces,

$$\begin{aligned}
\left|\prod_{k=1}^{n+1} a_k - \prod_{k=1}^{n+1} b_k\right| &= \left|\left(\prod_{k=1}^{n+1} a_k - a_{n+1}\prod_{k=1}^{n} b_k + a_{n+1}\prod_{k=1}^{n} b_k - \prod_{k=1}^{n+1} b_k\right)\right| \\
&= \left|a_{n+1}\left(\prod_{k=1}^{n} a_k - \prod_{k=1}^{n} b_k\right) + \left(\prod_{k=1}^{n} b_k\right)(a_{n+1} - b_{n+1})\right| \\
&\leqslant |a_{n+1}|\left|\prod_{k=1}^{n} a_k - \prod_{k=1}^{n} b_k\right| + \prod_{k=1}^{n} |b_k|\,|a_{n+1} - b_{n+1}| \\
&\overset{\substack{a_k, b_k \in \overline{\mathbb{D}} \\ \downarrow}}{\leqslant} \left|\prod_{k=1}^{n} a_k - \prod_{k=1}^{n} b_k\right| + |a_{n+1} - b_{n+1}| \leqslant \sum_{k=1}^{n} |a_k - b_k| + |a_{n+1} - b_{n+1}|\,.
\end{aligned}$$

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